Disease Risk Prediction Using Machine Learning

Team Information

- **Group**: G2

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Problem Description

Machine learning is widely used in healthcare to analyze patient data and assist in early disease prediction. Many diseases such as heart disease can be predicted by studying clinical parameters like age, blood pressure, cholesterol level, blood sugar, heart rate, and other medical attributes. Manual analysis of large patient records is difficult and time-consuming, so an automated system can be very useful.

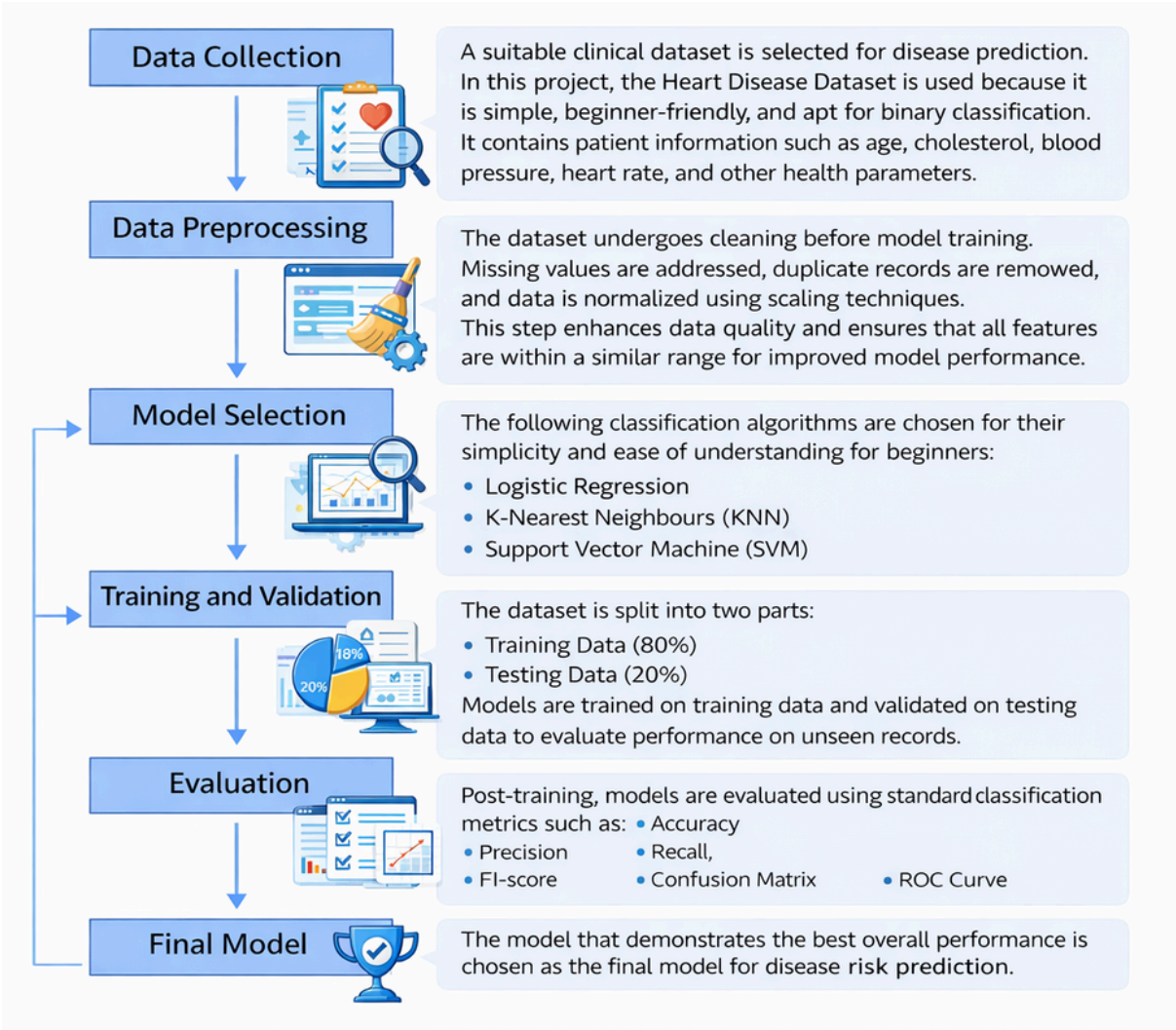
The objective of this project is to build a simple machine learning classification system that predicts whether a patient is at risk of disease or not using clinical data. The project uses a healthcare dataset and applies preprocessing techniques such as data cleaning and normalization to improve data quality. Three machine learning algorithms—Logistic Regression, K-Nearest Neighbours (KNN), and Support Vector Machine (SVM)—are used for training and testing.

The models are evaluated using accuracy, precision, recall, F1-score, confusion matrix, and ROC curve. This project demonstrates a complete beginner-level machine learning pipeline and shows how AI can support decision-making in the healthcare field.

Steps of Implementation

This project adheres to a straightforward machine learning pipeline. Each phase of implementation is outlined below:

1. **Data Collection**
   A suitable clinical dataset is selected for disease prediction. In this project, the Heart Disease Dataset is used because it is simple, beginner-friendly, and apt for binary classification. It contains patient information such as age, cholesterol, blood pressure, heart rate, and other health parameters.
2. **Data Preprocessing**
   The dataset undergoes cleaning before model training. Missing values are addressed, duplicate records are removed, and data is normalized using scaling techniques. This step enhances data quality and ensures that all features are within a similar range for improved model performance.
3. **Model Selection**
   The following classification algorithms are chosen for their simplicity and ease of understanding for beginners:
   - Logistic Regression
   - K-Nearest Neighbours (KNN)
   - Support Vector Machine (SVM)
4. **Training and Validation**
   The dataset is split into two parts:
   - Training Data (80%)
   - Testing Data (20%)
5. Models are trained on training data and validated on testing data to evaluate performance on unseen records.
6. **Evaluation**
   Post-training, models are evaluated using standard classification metrics such as:
   - Accuracy
   - Precision
   - Recall
   - F1-score
   - Confusion Matrix
   - ROC Curve
7. **Final Model**
   The model that demonstrates the best overall performance is chosen as the final model for disease risk prediction.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18%
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• Precision • Recall,
• FI-score • Confusion Matrix • ROC Curve
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Expected Modules / Libraries to be Used

The following Python libraries and key Scikit-learn modules are anticipated for use in this project:

1. **NumPy**
   Utilized for numerical operations and efficient handling of arrays in Python.
2. **Pandas**
   Used for loading the dataset from CSV files and for organizing, cleaning, and manipulating tabular data.
3. **Matplotlib**
   Employed for plotting graphs such as ROC curves and performance visualizations.
4. **Seaborn**
   Primarily used for advanced data visualization, such as drawing the confusion matrix heatmap.
5. **Scikit-learn (sklearn)**
   The core machine learning library used for preprocessing, model training, testing, and evaluation.

Important Modules from Scikit-learn

- **train_test_split**: Divides the dataset into training and testing data.
- **StandardScaler**: Normalizes or scales input features to ensure they are within a similar range.
- **LogisticRegression**: Implements the Logistic Regression classifier.
- **KNeighborsClassifier**: Implements the K-Nearest Neighbours (KNN) classifier.
- **SVC**: Implements the Support Vector Machine (SVM) classifier.
- **accuracy_score**: Calculates model accuracy.
- **precision_score**: Computes model precision.
- **recall_score**: Determines model recall.
- **f1_score**: Calculates the model's F1-score.
- **confusion_matrix**: Generates the confusion matrix.
- **roc_curve**: Creates the ROC curve for model evaluation.

Expected Evaluation Metrics and Plots to be Used

The following evaluation metrics and plots will be employed to compare machine learning model performance:

1. **Accuracy**
   Measures the correctness of predictions made by the model out of the total predictions.
2. **Precision**
   Assesses how many predicted positive cases are actually correct, crucial for reducing false positives.
3. **Recall**
   Determines the number of actual positive cases correctly identified by the model, which is vital in medical contexts where missing a disease case can be risky.
4. **F1-Score**
   Provides a balanced measure of model performance as the harmonic mean of precision and recall.
5. **Confusion Matrix**
   Displays the number of correct and incorrect predictions, aiding in understanding true positives, true negatives, false positives, and false negatives.
6. **ROC Curve**
   Used to compare model classification performance at different thresholds.
7. **AUC Score (Optional but Recommended)**
   Indicates how well the model can separate classes; a higher AUC suggests better performance.

Plots to be Used

- Confusion Matrix Heatmap
- ROC Curve Plot
- Model Comparison Bar Graph (Accuracy / F1-score)

Required Topics from the Subject

1. **Strings**
   Used for managing text data, handling column names, labels, file paths, and displaying output messages.
2. **Pointers**
   Though primarily a concept in C programming, understanding pointers aids in efficient data handling and memory management.
3. **Files**
   Essential for reading and writing data, as the clinical dataset is stored in a CSV file and loaded into Python for processing.
4. **Arrays**
   Used to store multiple values of the same type, crucial for storing numerical feature values and model input data.
5. **Functions**
   Help divide the program into smaller, reusable blocks. Useful for preprocessing, training models, and evaluating performance.
6. **Conditional Statements**
   Facilitate decision-making in a program, such as checking for missing values or selecting the best model based on accuracy.

Platform Required

The project can be implemented using the following platform:

- **Jupyter Notebook**: Ideal for writing and executing Python code incrementally. It is user-friendly and well-suited for machine learning experiments.

Books and Link Sources

- Scikit-learn Documentation – https://scikit-learn.org
- Pandas Documentation – https://pandas.pydata.org
- NumPy Documentation – https://numpy.org
- Matplotlib Documentation – https://matplotlib.org

Dataset-

Repo-*https://github.com/SnehilJoshi007/PBL-AIML.*

📊 Dataset Overview

Parameter	Details
Purpose	Synthetic dataset for disease risk prediction using patient clinical data. Designed for data cleaning, preprocessing, feature selection, and machine learning classification.
Total Rows	1800
Total Columns	12
Data Quality Design	Includes missing values (~6%) and some noise/outliers to simulate real-world clinical data inconsistencies.

📁 Column Details

Column	Description	Typical Unit	Missing Values (~)
Age	Patient age	Years	~108
Gender	0 = Female, 1 = Male	Binary	~108
BMI	Body Mass Index	kg/m^2	~108
Blood_Pressure	Systolic blood pressure	mmHg	~108
Cholesterol	Serum cholesterol level	mg/dL	~108
Glucose	Blood sugar level	mg/dL	~108
Heart_Rate	Heart beats per minute	bpm	~108
Smoking	Smoking status	Binary	~108
Alcohol_Intake	Weekly alcohol consumption	Units/week	~108
Physical_Activity	Weekly exercise duration	Hours/week	~108
Family_History	Genetic predisposition	Binary	~108
Disease_Risk	Target variable (0 = Low, 1 = High)	Binary	~108

⚡ Quick Summary

Column	Min	Max	Comment
Age	18	79	Wide age distribution across adults
BMI	~15	~40+	Includes underweight to obese cases
Blood Pressure	90	180+	Contains hypertension cases
Cholesterol	150	300+	Some high-risk values present
Glucose	70	190	Includes diabetic range
Heart Rate	55	115	Includes abnormal heart rates
Physical Activity	0	10	Covers sedentary to active lifestyles
Disease Risk	0	1	Balanced classification target

A	B	C	D	E	F	G	H	I	J	K	L
Age	Gender	BMI	Blood_Pressure	Cholesterol	Glucose	Heart_Rate	Smoking	Alcohol_Intake	Physical_Activity	Family_History	Disease_Risk
63	1	25.1	113	219	124	62	1	8	8.3	1	
	0	25.1	112	150		78	0	3	9.9	1	0
46	0	28.9	111	219	105	86	1	6	0.5	0	
52	1	26.4		215	184	85	0	0	2.1	0	1
56		30.4	168	254	178	74	0	6	2.1	1	1
35	0	27.2	111	191	162	109	1	1	3.8	1	1
37	1	29.4	99	283	152	69	1	6	6.5	1	1
	0	28	137	274	162	76	1	7	7.7	0	1
77	0	28.9	151	251	152	69	0	3	3.4	1	1
75	0	25.8	158	180	143	99	1	3	3.1	1	0
40	0	27	99	187	185	109	0	1	1.8	0	0
51	0	24.8	124		98	113	0	1	2	1	1
50	1	28.7	110	156	115	87	0	9	5	0	0
67	0	19.9	100	273	126		1	7	5	1	0
65	0	29.3	92	165	122	84	1	10	5.1	1	
27	1	30.5	176	168	108	75	0	10	6.7	0	1
50	1	29.3	139	298		96	1	8	6.1	0	1
64	0	25.5	180	208	122	112	0	5	3.2	1	1
50	0	28.3	176	233	73	71	1	9	0.9	0	1
65	1	22.5		241	123		1	8	2.4		1
	1	29.8	132	241	142	72	0	4	2.6	1	
37	1	21.1	98	280	139	85	0	2	2.7	0	
32	1	28.7	93	262	81	65	1	5	7.9	1	1
79		28.4	98	219	152		1	3	1.2	0	